EXPERIMENT No.: 3

AIM OF EXPERIMENT: Interfacing of suitable sensor and actuator to DAQ - Demonstration of bottle filling plant using relay system.

OBJECTIVE :
1. To study various steps involved in design of bottle filling plant.
2. To verify the operation of the bottle filling plant.
3. To study construction and working of relay.

APPARATUS : Bottle filling plant trainer model, proximity sensor, dc motor, HMI, relay switches.

INTRODUCTION :

In mass manufacturing process particularly bulk liquids, medicines, food products, mineral water etc. packing & dispatches is crucial job which involves lot of different parameters directly affecting time factor & production cost. Depending on type of liquid, packing size, type of containers we have to design filling plant & control algorithm.

There are two common types of setups.

1] Conveyor belt system & 2] Rotating table system.

Conveyor belt system is suitable when containers are filled one by one, one at a time at input stage of filling plant. This system is further automated for automatic moments of containers to packing crets & cartons.

It is necessary to design a plant compatible for metallic and non metattlic devices. It consists of conveyor, switches, proximity sensor to sense, Proximity sensor inductive type to sense metal and non metal .

Tank contains liquid to fill with motor to up it and filter with it.

1. **CONVEYOR :**

   It is a belt connected to chassis of an AC servomotor and it rotates with rotation of motor. An object is to be placed on the belt then it is moved towards filling section using the conveyor mechanism. A Flat Belt Conveyor as is used to support the

bottles on their location & carry them forward sequentially to fill with water. A Flat Belt is passed over three rollers to performed rolling action.

2. **PROXIMITY SWITCHES**

   To confirm that object is properly placed in a perfect position or not, a proximity switch is used. It indicates the appearance of the object in front of them. For metallic object inductive switches are used and for non-metallic object capacitive switches are used.

3. **FILLER :**

   Filler use tap and AC pump assembly to pass the liquid forward bottles.

4. **OUTPUT INDICATOR:**

   LED are used for indication of different operations such as filling fault, metals , non – metals etc.

5. **LOAD SENSING DEVICES:**

   In case if partially filled bottles comes on belt that is sensed by limit switch and in turn fault LED indicated and bottle is passed as it without filling.

6. **RELAY**

   A relay as shown in figure 6 is an electrically operated switch. Many relays use an electromagnet to mechanically operate a switch, but other operating principles are also used, such as solid state relays. Relays are used where it is necessary to control a circuit by a low-power signal (with complete electrical isolation between control and controlled circuits).

7. **ROTARY PUMP**

   A Rotary operated pump arrangement as shown in figure uses a rotary pump arrangement to provide a fixed volume of liquid to flow into the bottle according to time based on user input.

BOTTLE FILLING PLANT:

Bottle filling plants are occupying a big sector of the various business circles now – days. As there is great competition in water business, the bottle filling plant model can help the water plant to increase their productivity.

This model is generally used for the filling of liquid (like water and other liquid beverages) into 300 ml to 500 ml of metallic and non-metallic bottles of any shape by changing the items time of the timer inbuilt in PLC. It is possible to fill different volumes time count in the timer, by using word accurate it means that the liquid is delivered exactly into the bottle without spillage of the liquid outside the bottle. The operating parameters such as quality of the product momentum of the bottle conveyor filling head opening and filling speed for different bottle sizes can be taken care of the PLC interface with the plant model. Auto protective features are also introduced in the model. Guide plates are provided to check the alignment and orientation of the bottles passing on conveyor.

The empty bottles are loaded on to the conveyor or positioned individually on conveyor manually. With both the side bearings one acts as driver and other driven. The conveyor width is about the conveyor bed belt. An A. C. motor drives the driving system. The bottles enter the filling area continuously on the conveyor with certain time gap between them. They are sensed by proximity sensors and ensure smooth and trouble less operation.

The proximity sensors used in this model can also be used for position sensing of the bottle.

Filling unit mainly consists of submersible pump and limit switch as a weight sensor. This unit starts when the bottles are positioned correctly under filling head thus reducing any over or under fill and also virtually eliminated any spillage of liquid on to conveyor.

After bottle gets filled, limit switch gets operated and the conveyor moves and allows next bottle to get into the position. The filling bottles are connected at unloading end of the conveyor which is manually removed. Here is the model it requires manually intervention while placing the bottle on the conveyor and removing the bottle from the conveyor. Hence the bottle filling plant here is semiautomatic one.

PROCESS:

1. Place the bottle on conveyor.
2. Start the motor.
3. When proximity sensor will sense the bottle, stop the motor.
4. Start the pump and fill the bottle till limit switch output is on.
5. Stop the pump and start the motor.
6. Collect the bottle when it will reach at the end.

CONCLUSION :

Demonstration of bottle filling plant using relay system is completed which included : various steps involved in design of bottle filling plant , verify the operation of the bottle filling plant, study construction and working of relay.

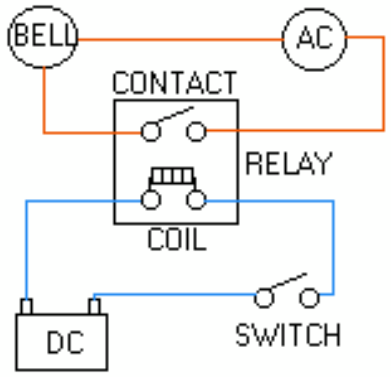


Fig. : Relay system

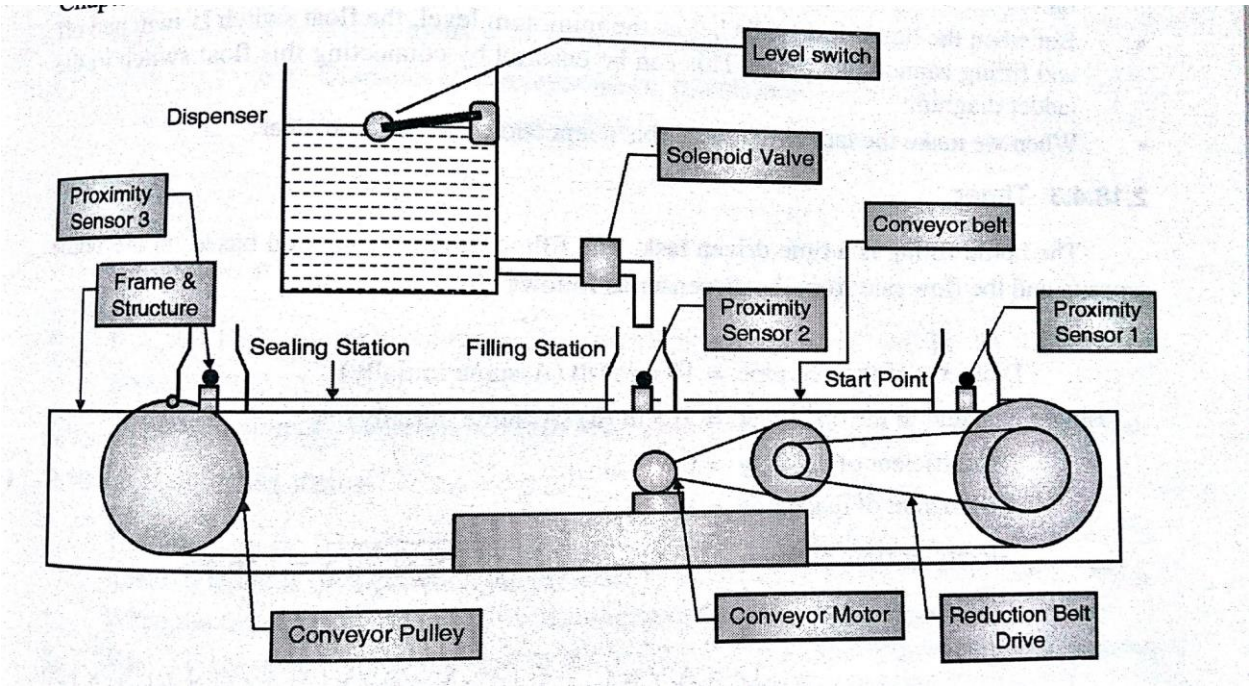


Fig. : Bottle filling station.